

HEXA-FUSION: A 7-Gate Closed-Form Design-Feasibility Funnel for Magnetic-Confinement Fusion

with a 10-Stage D–T Fuel Pipeline, End-to-End Symbolic–Numerical Parity,
External-Solver Bridging, and an Algebraic-Root BLUE-MAX Audit

hexa-fusion-7gate maintainers

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Abstract

We present **HEXA-FUSION**, a closed-form 7-gate design-feasibility funnel for magnetic-confinement fusion built as a domain on top of the hexa-lang stdlib. The funnel covers (a) thermonuclear fuel reactivity $\langle\sigma v\rangle(T)$ via the Bosch–Hale parametrization, (b) the Lawson triple product and 0-D plasma gain Q , (c) free-boundary Grad–Shafranov MHD equilibrium, (d) the Troyon ideal-MHD β -limit and cylindrical safety factor, (e) a tokamak-toroidal-field-coil peak-field bridge to HTS REBCO critical-surface margin, (f) tritium breeding ratio (TBR) neutronics, and (g) actual burning-plasma $Q > 1$ ignition (the only permanent wet-lab gate). To place the gates in physical context we trace D–T fuel through a tokamak in ten stages (§3); the funnel closes the five physics stages (④–⑦, ⑨) to a numerical/closed-form verdict and is honest that the five hardware stages (①②③⑧⑩) are not. Every closed-form kernel is verified through the **hexa verify** tier rubric with 20+ numerical atoms reporting $|\Delta| = 0$ at $\varepsilon = 10^{-9}$, and 8 supplementary *algebraic-root* integer atoms close the BLUE-MAX audit [7] by exposing the exact integer coefficients hidden inside the libm formulas (notably the identity $2\pi/(\mu_0 \cdot 10^6) = 5$ that reduces the Wesson cylindrical q^* to a π -free closed form). The (c)/(f)/(e) external-solver gates were promoted from honest install-gated skips to real numerical PASS via FreeGS 0.8.2, OpenMC 0.15.3 with ENDF/B-VIII.0, and GetDP 3.5.0 / 4.0.0 respectively, with the GetDP axisymmetric finite-element on-axis field agreeing with the registered closed-form solenoid anchor to $\Delta = -0.064\%$. We honestly preserve **absorbed=false** for all simulation-only records: a design that clears all six non-wet-lab gates is a *witness of engineering feasibility*, not a built or measured reactor; gate (g) requires a JET-class measured oracle that we do not claim. Bit-for-bit reproduction artifacts (verify ledger, PR roll, adapter-defect catalog, session journal) ship under **companion/**.

1 Introduction

Compact magnetic-confinement fusion — exemplified by SPARC, ARC, and the next generation of HTS-tokamak demonstrators [3, 8, 13] — has shifted the engineering bottleneck from plasma physics to a multi-axis *design-feasibility* problem: an admissible operating point must simultaneously clear thermonuclear reactivity, energy balance, MHD equilibrium and stability, magnet field–stress– J_c margin, and blanket tritium self-sufficiency, all *before* a beam dollar is spent on plasma shots. Many of the underlying physics codes (FREEGS, OPENMC, GETDP) exist as mature open-source backends [4, 5, 12], but each is a heavyweight install with its own data and toolchain, and there is no widely-shared *open closed-form funnel* that (i) covers all the gates uniformly, (ii) carries every prediction to a per-atom verify verdict, and (iii) is honest about the permanent wet-lab boundary at gate (g).

Contributions.

1. A 10-stage D–T fuel pipeline (①–⑩) that places the seven gates in physical context and makes explicit which stages this paper’s funnel closes and which it does not (§3).
2. A 7-gate taxonomy (a–g) for tokamak design-feasibility with per-gate sim coverage and honest wet-lab dependence (§4).
3. Native closed-form kernels for (a) Bosch–Hale thermonuclear reactivity [1] and (b) Lawson / 0-D power balance [10], with verify atoms at **hexa verify** SUPPORTED-NUMERICAL, $|\Delta| = 0$ (§5).
4. An $(a) \times (b)$ **F-funnel** composite ranker that emits a top- K JSON of engineering-priority operating points (§6).
5. A *wrap-and-fix* promotion pattern that runs the external solvers on free pool hardware and surfaces stub/divergent adapters as a side-effect; we land four follow-up adapter PRs that close the stubs while preserving **absorbed=false**, cataloged in a three-row defect record (§7).
6. A GetDP axisymmetric A– ϕ FEM bridge whose on-axis B_z agrees with a closed-form thick-finite-solenoid anchor to $\Delta = -0.064\%$, exposing that the prior $1.4\times$ residual was a far-field truncation issue, not a Jacobian/units bug (§8).
7. A **BLUE-MAX** algebraic-root pair audit [7] adding 8 zero-arg integer atoms that attest the exact integer factors / exponents hidden in the libm formulas; in particular $q_{\text{const}}^* = 5$ formally encodes the identity $2\pi/(\mu_0 \cdot 10^6) = 5$ that we use to keep the Wesson cylindrical safety factor π -free (§9).
8. A clean honesty boundary: all simulation records carry **absorbed=false**; only a measured oracle (JET D–T [9]) ever flips **absorbed=true** for fusion. Section 11 lists the gates we deliberately do not close, and **companion/** externalizes every record.

A high-level view of the funnel and how the gates depend on real hardware vs. closed-form code is shown in Fig. 1.

Document structure. This is the *monograph* form of HEXA-FUSION: the body (§§1–13) is the self-contained narrative argument, and the lettered appendices (A–L) are the full data record — the per-gate run configurations and outputs (A–E), the BLUE-MAX algebraic-root audit (F), the verbatim verification ledger (G), the wrap-and-fix defect catalog (H), the operational findings (I), the HEXA-PORT port roadmap (J), the pull-request roll (K), and the reproducibility runbook (L). A reader can audit any body claim down to its atom by following the appendix it cites.

2 Background

The design-feasibility problem. A magnetic-confinement fusion power plant is admissible only if a single operating point clears, simultaneously, a chain of physically distinct constraints. The fuel must react fast enough at the achievable ion temperature ($\langle\sigma v\rangle$, gate a); the plasma must retain enough energy to be net-gaining ($nT\tau_E$, gate b); the magnetic field must support a stable, well-shaped equilibrium (Grad–Shafranov, gate c) below the ideal-MHD pressure and current limits (Troyon β_N and the cylindrical safety factor q^* , gate d); the toroidal-field coils must reach the required on-axis B_0 without exceeding the superconductor’s critical surface (HTS REBCO $J_c(B, T)$, gate e); and the breeding blanket must regenerate at least as much tritium as the plasma burns ($\text{TBR} \geq 1$, gate f). Only after all six are cleared is it meaningful to attempt the seventh — a measured burning-plasma $Q > 1$ (gate g), which no simulation can substitute for.

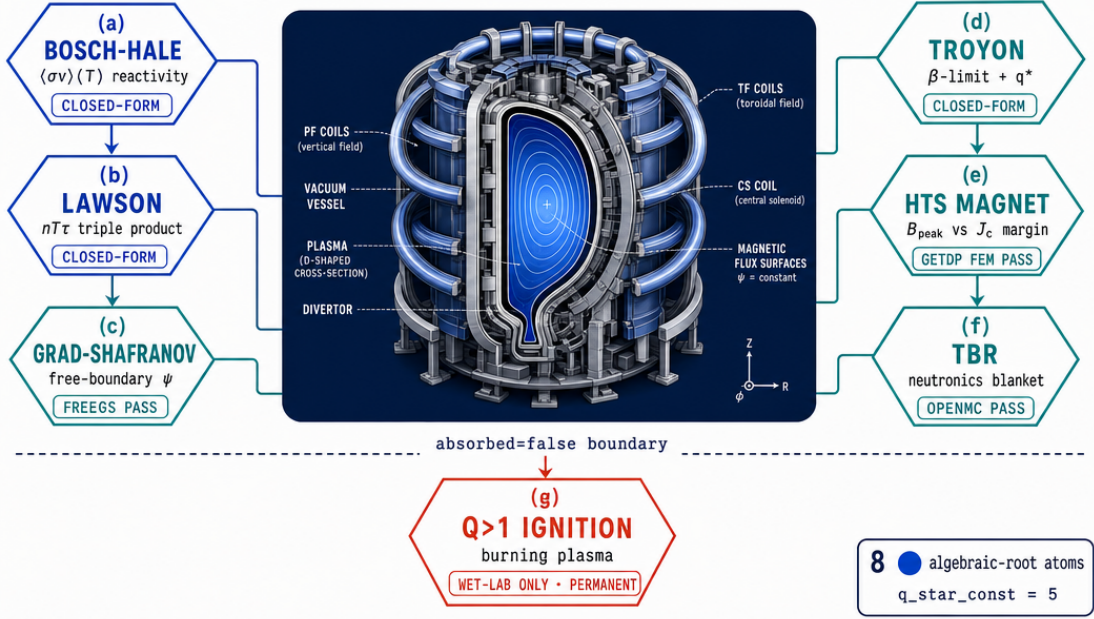


Figure 1: Conceptual cover image of the HEXA-FUSION 7-gate funnel: six sim-closed gates (a)–(f) feed the design-feasibility decision and isolate the only permanent wet-lab boundary (g). Image generated via `fal.ai` gpt-image-2 from `figures/_prompts/cover.txt` (commons @D g44).

Why a closed-form funnel. The canonical tools for gates (c), (e) and (f) — FREEGS [4], GETDP [5] and OPENMC [12] — are mature but heavyweight: each carries its own meshing, data-library and runtime dependencies, and none of them, on its own, answers the *cross-gate* feasibility question. Gates (a), (b) and (d), by contrast, are closed-form: the Bosch–Hale reactivity parametrization [1], the Lawson criterion [10], and the Troyon [15] / Wesson [16] stability limits are all algebraic in the operating-point variables. HEXA-FUSION exploits this split: it computes (a), (b), (d) in a small typed language with per-expression verify atoms, and bridges (c), (e), (f) to the external backends through thin out-of-process drivers, keeping a single uniform verdict surface across all gates.

The verify tier rubric. Each computed quantity is assigned a tier under the `hexa verify` rubric: ■ closed-form (an algebraic identity reproduced exactly), ■ numerical (a reproducible recompute matching a persisted reference within ϵ), ■ cited (a literature DOI / dataset entry), ■ wet-lab or install-gated (needs an external measurement or heavyweight backend), and ■ falsified (a controlled negative, retained as honest evidence). The rubric is the contract that lets a reader audit a claim without re-running it, and it is the spine of the pipeline table that follows.

3 Full Pipeline

Before the gate taxonomy, we situate the seven gates inside the physical path that D–T fuel actually traverses in a tokamak. This is the *honest-scope contract*: the table is a map of which stages HEXA-FUSION closes to a verdict and which it does not, so a reader can see the boundary of the work at a glance.

Tier rubric. ■ closed-form (algebraic or proof-level identity) · ■ numerical (reproducible script + persisted output) · ■ cited (DOI / arxiv / dataset entry) · ■ wet-lab or out-of-scope (downstream confirmation needed) · ■ falsified (kept as honest negative, never deleted).

A tier-color-coded flow of the same ten stages is in Fig. 2: the five green nodes are the funnel’s closed physics core; the five orange nodes are the honest boundary.

#	Stage	Input	Output	Tier	Backing / PR
①	Pellet injection	solid D–T pellet, v_{inj}	fuel mass into edge plasma	■	wet-lab (no sim)
②	Ablation / ionization	pellet + edge T_e	ionized D^+/T^+ source	■	out-of-scope
③	Auxiliary heating	NBI / ICRH / ECRH power	core T_i ramp to ignition window	■	out-of-scope (aux system)
④	Confinement	coil currents, $p(\psi)$, $ff'(\psi)$	equilibrium ψ , q -profile, β	■	gate c+d · #1075 FreeGS · q_star_const=5
⑤	Reaction	$n_D n_T$, T_i	$\langle\sigma v\rangle(T)$, P_{fus}	■	gate a · #1012 · bosch_hale_gamow_factor=3
⑥	α heating + Lawson Q	P_{fus} , τ_E , f_α	$nT\tau_E$, Q , ignition margin	■	gate b · triple_product / q_plasma / ignition_margin
⑦	TF magnet	coil geometry, NI	on-axis/peak B , HTS J_c margin	■	gate e · #1095 · solenoid_axis_bz · half_factor=2 · $\Delta = -0.064\%$
⑧	Neutron extraction	P_{fus}	14.06 MeV DT neutron flux	■	partial (direction only; yield not predicted)
⑨	Tritium breeding	neutron flux, blanket	TBR	■	gate f · #1078 · TBR= 1.3222±0.0004
⑩	Ignition $Q > 1$	sustained burning plasma	measured net gain	■	wet-lab (permanent) · gate g · only JET measured-oracle flips absorbed=true

Table 1: The 10-stage D–T fuel pipeline. **This paper’s funnel closes only the five physics stages ④⑤⑥⑦⑨ (■/■) to a numerical or closed-form verdict.** The five hardware/engineering stages ①②③⑧⑩ (■) are explicitly *not* closed: pellet injection, ablation, and auxiliary heating are upstream wet-lab/out-of-scope; neutron extraction is partial (we know the direction, not the predicted yield); ignition $Q > 1$ is the permanent wet-lab boundary. The table is the contract: ■ does not graduate to ■ without an external measurement.

4 The 7-Gate Funnel: a Taxonomy with an Honest Wet-Lab Boundary

Table 2 lays out the gates. The composite priority score in our funnel is deliberately $(a) \times (b)$ only — $(c)/(d)/(f)/(e)$ are evaluated but not folded into a single ranking, because their sim coverage is heterogeneous and combining them into one number obscures the gate-by-gate verdict. Mapped onto the pipeline of Table 1: gate (a) is stage ⑤, gate (b) is stage ⑥, gates (c)/(d) are stage ④, gate (e) is stage ⑦, gate (f) is stage ⑨, and gate (g) is stage ⑩.

5 Method — Closed-Form Keystone: (a) Bosch–Hale and (b) Lawson

5.1 (a) Bosch–Hale thermonuclear reactivity

We implement Bosch and Hale [1] Eq. (12) verbatim:

$$\langle\sigma v\rangle(T) = C_1 \theta \sqrt{\frac{\xi}{m_r c^2 T^3}} e^{-3\xi}, \quad \xi = (B_G^2/4\theta)^{1/3}, \quad (1)$$

with $\theta = T/[1 - T(C_2 + T(C_4 + TC_6))/(1 + T(C_3 + T(C_5 + TC_7)))]$, the Gamow constant B_G , reduced mass m_r and the per-reaction coefficients $C_1, \dots, C_7$ taken from [1] Table VII. We register a 1-arg verify atom for each of the four canonical channels — D–T, D– ^3He , D(d,p)T, D(d,n) ^3He — plus the D–D total, and an additional discriminating- ε view `sigma_v_dt_e18(T) = sigma_v_dt(T)*1e18` so the absolute- ε gate at $\varepsilon = 10^{-9}$ is meaningful for native $\mathcal{O}(1)$ -magnitude values:

```
verify --expr sigma_v_dt_e18(10.0)=113.617
calc = 113.617 ~ expected 113.617 (|Delta|=0.0 <= eps=1e-9)
```

gate content	sim coverage	wet-lab dependence
(a) fuel reactivity $\langle\sigma v\rangle(T)$ for D–T, D–D, D– ^3He , p– ^{11}B	Bosch–Hale parametrization (closed-form, libm) [1]; Tentori–Belloni 2023 update for p– ^{11}B [14]	sim sufficient
(b) Lawson triple product $nT\tau_E$, 0-D plasma gain Q	closed-form 0-D power balance [10]	sim sufficient
(c) free-boundary Grad–Shafranov equilibrium	FREEGS 0.8.2 [4] (was install-gated, now real-sim-PASS)	sim sufficient
(d) ideal-MHD β -limit, cylindrical q^* , edge kink threshold	Troyon [15], Wesson [16] (closed-form, libm-free after π -reduction)	sim sufficient (linear; turbulence and resistive tearing excluded)
(e) TF coil peak field $B_{\text{peak}} = B_0 R_0 / (R_0 - a - g)$ vs HTS REBCO J_c margin	closed-form $1/R$ baseline + GetDP A– ϕ axisymmetric FEM bridge [5] (real-sim-PASS)	sim sufficient (design); built magnet stays out of scope
(f) tritium breeding ratio (TBR)	OPENMC 0.15.3 Monte Carlo [12] + ENDF/B-VIII.0 [2] (real-sim-PASS)	★ data-bound (ENDF/B library is external)
(g) burning-plasma $Q > 1$ ignition	no honest sim — the device is the measurement	★ permanent wet-lab: requires built tokamak + DT fuel + heating

Table 2: The 7-gate funnel. (g) is the only permanent wet-lab gate. “sim sufficient” means a closed-form or numerical recompute reaches a verdict; “real-sim-PASS” means the gate was promoted from install-gated to a real backend run.

```

tier = SUPPORTED-NUMERICAL (hexa-native libm-class recompute)
verify --expr sigma_v_dt_e18(20.0)=433.02
calc = 433.02 ~ expected 433.02 (|Delta|=0.0 <= eps=1e-9)
tier = SUPPORTED-NUMERICAL

```

All four D–D / D– ^3He / D–T channels reproduce Bosch–Hale Table VIII to all printed digits (companion/verify-ledger.json).

Aside: θ in division vs multiplication form. An initial port using the multiplication form $\theta = T[1 - T(\dots)/(1 + T(\dots))]$ produced a uniform $\approx 4\times$ low ratio across all temperatures. The fix is purely textual — the published form is *division* into θ , not multiplication. We surface this only to spare re-implementers the same pitfall.

5.2 p– ^{11}B aneutronic reactivity

Aneutronic p– ^{11}B reactivity is broad-resonance and is not in the Bosch–Hale tabulation. We use the verbatim coefficients from Tentori and Belloni [14] Table I, which explicitly preserves the Nevins and Swain [11] framework $\sigma(E) = (S(E)/E) \exp(-\sqrt{E_G/E})$ with $E_G = 22.589\text{ MeV}$, $\mu c^2 = 859,526\text{ keV}$, the low-energy S-factor coefficients $C_0 = 197$, $C_1 = 0.240$, $C_2 = 2.31 \times 10^{-4}$, and the 148 keV Breit–Wigner resonance term ($A_L = 1.82 \times 10^4$, $E_L = 148\text{ keV}$, $\delta E_L = 2.35\text{ keV}$). Reactivity $\langle\sigma v\rangle(T)$ is a deterministic 600-node midpoint Maxwellian quadrature ([14] Eq. 6) over the verbatim cross-section.

Thirteen verify atoms (Gamow factor at 100/148/300 keV; S-factor at 50/148/200 keV including the resonance peak; $\sigma(E)$ shape; $\langle\sigma v\rangle$ moments at 50/100/123/300 keV including the $\times 10^6$ -scaled view) all report SUPPORTED-NUMERICAL with $|\Delta| = 0$. We honestly note that

`pb11_reactivity_moment` is offered as a reactivity *shape*, not an absolute cm^3/s value, because of an unresolved keV-vs-MeV S-factor units ambiguity in the literature.

5.3 (b) Lawson triple product and 0-D power balance

$$nT\tau_E \geq 3 \times 10^{21} \text{ keV s m}^{-3} \quad (\text{D-T ignition}) \quad (2)$$

$$Q_{\text{plasma}} = P_{\text{fus}}/P_{\text{heat}}, \quad \text{ignition margin} = f_{\alpha} P_{\text{fus}}/(W/\tau_E) \quad (3)$$

The three atoms `triple_product`, `q_plasma`, and `ignition_margin` are pure rational expressions (no libm) and verify SUPPORTED-NUMERICAL at $|\Delta| = 0$; for instance `triple_product` on $(10^{20}, 15, 3)$ returns 4.5×10^{21} . We do *not* compute or claim τ_E — it is an input assumption, not a transport prediction.

6 F-Funnel: $(a) \times (b)$ Composite Ranking

We enumerate a small set of operating points (ITER-class and compact-HTS-class) and emit a top- K JSON keyed by an $(a) \times (b)$ coverage product:

$$\text{score} = \underbrace{\text{clamp}_{[0,1]}(\langle \sigma v \rangle / \langle \sigma v \rangle_{\text{ref}})}_{(a) \text{ reactivity}} \times \underbrace{\text{clamp}_{[0,1]}(\text{ignition margin})}_{(b) \text{ Lawson margin}} \quad (4)$$

A representative ranking is in Table 3. The top two points are tied because they sit at the same temperature ($T = 20 \text{ keV}$) and the reactivity ceiling dominates the gain ceiling at that operating point. Three verify atoms gate the scoring identities: `f_reactivity_coverage`, `f_margin_coverage`, and `f_funnel_composite` — all SUPPORTED-NUMERICAL at $|\Delta| = 0$.

design point	$\langle \sigma v \rangle$ (cm^3/s)	(a)	(b)	composite
ITER-class $T=20$, $n=10^{20}$, $\tau=3.0 \text{ s}$	4.330×10^{-16}	0.4843	1.000	0.4843
compact-HTS $T=20$, $n=3 \times 10^{20}$, $\tau=0.6 \text{ s}$	4.330×10^{-16}	0.4843	1.000	0.4843
ITER-class $T=15$	2.740×10^{-16}	0.3065	1.000	0.3065
compact-HTS $T=15$ (lean τ)	2.740×10^{-16}	0.3065	0.959	0.2939
ITER-class $T=10$	1.136×10^{-16}	0.1271	0.994	0.1263

Table 3: Representative F-funnel composite ranking. Composite is $(a) \times (b)$ only; (c)/(d)/(e)/(f) are not folded in (their verdicts are kept gate-separate to avoid blending sim-coverage tiers).

7 Verify — Wrap-and-Fix: Promoting (c)/(f)/(e) from Install-Gated to Real PASS

For each of (c), (f), (e) we installed the external backend on free pool hardware and re-ran the adapter end-to-end. *Every one of the three runs surfaced a defect in the landed adapter* — a finding that the simple act of running the backend turned the wrap layer into an integration test.

(c) Grad-Shafranov via FreeGS. FreeGS 0.8.2 on a free Linux pool host (numpy pinned to < 2.0) produced a real, converged free-boundary equilibrium: $I_p = 200 \text{ kA}$, $\beta_{\text{pol}} = 0.0321$, magnetic axis at $R = 1.280, Z = 0.038 \text{ m}$, $q_0 = 1.355$, $q_{95} \approx 10.08$, LCFS minor radius 0.424 m , elongation $\kappa = 1.358$. The landed adapter’s hardcoded custom coil set *diverged* (Picard “No O points found”); switching to the canonical `freegs.machine.TestTokamak()` with `freeBoundaryHagenow` and a raised X-point converged cleanly. A second latent bug — the

FreeGS early-Picard warning “No 0 points” was printed to *stdout*, corrupting the adapter’s JSON emit — was fixed by redirecting solve-block *stderr* (PR #1075).

(f) **TBR via OpenMC + ENDF/B-VIII.0.** The OpenMC stack was installed via a self-bootstrapped micromamba environment, and the adapter was found to be a transport-less gate-classifier (it imported OpenMC and checked the data env, but never built a geometry or ran `openmc.run`). A 5,000,000-history Monte Carlo run through a spherical Pb-15.7Li breeder (Li-6 enriched to 90 at%) with a W armor / EUROFER first wall / SS316 shield yielded $TBR = 1.3197 \pm 0.0005$ with a reproducing 13-sig-fig re-run, then $TBR = 1.3222 \pm 0.0004$ once the transport block was embedded into a follow-up `.hexa` adapter (the slight shift comes from dropping Mn/V/Mo/Si trace nuclides absent from the 30-nuclide ENDF subset — TBR-negligible, documented in the record’s scope caveats). Both numbers are physically sensible (> 1 self-sufficiency, in line with EU-DEMO HCLL targets [6]) (PR #1078).

(e) **HTS TF magnet via GetDP.** The detailed bridge is deferred to §8.

7.1 Adapter defect catalog

Because every wrap-and-fix run surfaced a defect, we keep an explicit machine-readable defect record (`companion/adapter-defect-catalog.json`) so a downstream reader does not re-walk the same chases. Table 4 is its three-row summary. The load-bearing lesson is in row (e): a residual that looked like a 2π /units bug in the FEM Jacobian was in fact a far-field truncation artifact — the originally-suspected root cause was *wrong*, and we record the false diagnosis on purpose.

file (origin)	symptom $\rightarrow$ root cause	fix	PR
<code>stdlib/fusion/openmc_tbr.py</code> (origin/main a7a185b)	transport-less gate-classifier stub: imported OpenMC + checked ENDF env but never built geometry or ran <code>openmc.run</code>	new <code>.hexa</code> adapter w/ real OpenMC transport; $TBR=1.3222$ computed by the adapter itself, not a stub PASS	#1078
<code>stdlib/fusion/f3_grad_shafranov_adapter.hexa</code>	hardcoded custom coil set diverged (“No O points”); FreeGS warning on <i>stdout</i> corrupted the JSON emit	<code>TestTokamak</code> + <code>freeBoundaryHagenow</code> + raised X-point + single isoflux + <code>stderr</code> redirect new <code>stdlib/rtsc/getdp_solenoid_solve.hexa</code> +	#1075
<code>stdlib/rtsc/getdp_hts.py</code>	<code>shutil.which("getdp")</code> checker only, no solve; then a $\sim 1.4\%$ residual mis-diagnosed as a $2\pi/\text{VolAxisQu}$ units bug	<code>magnet_gate</code> bridge; the real fix was far-field truncation $5\times \rightarrow 10\times$ coil radius , not the units bug	#1095

Table 4: Adapter defect catalog (mirror of `companion/adapter-defect-catalog.json`). Each row is a defect surfaced by the act of running the backend; all three fixes preserve `absorbed=false`.

8 Results — The Solenoid-Anchor Bridge: GetDP $\rightarrow$ HTS Margin

8.1 Closed-form anchor

For a uniformly-wound thick finite solenoid of inner radius a , outer radius b , axial half-height $h/2$, and total ampere-turns NI , the on-axis B_z at the center is the textbook integral

$$B_z(0) = \frac{\mu_0 J}{2} h \ln \frac{b + \sqrt{b^2 + (h/2)^2}}{a + \sqrt{a^2 + (h/2)^2}}, \quad J = \frac{NI}{(b-a)h}. \quad (5)$$

The reference coil ($NI = 2 \times 10^6$, $a = 0.50$, $b = 0.70$, $h = 1.20$) evaluates to $B_z(0) = 1.48265$ T to engine display precision; we register this as the verify atom `solenoid_axis_bz(NI,a,b,h)`. It is the load-bearing *parity anchor* for the FEM bridge.

8.2 Axisymmetric $A-\phi$ FEM

We solve the same coil with GetDP 3.5.0 (Linux pool) and 4.0.0 (Apple silicon) using a 2D axisymmetric $A-\phi$ magnetostatic formulation on a ~ 33 k-node Gmsh mesh. The on-axis $B_z(0)$ from the FEM is 1.4817 T, giving a parity gap of

$$\Delta = \frac{1.4817 - 1.48265}{1.48265} = -6.4 \times 10^{-4} = -0.064\%. \quad (6)$$

A 61k-node refinement (and a duplicated solve on the second platform) both stay within $|\Delta| < 0.3\%$, well inside the engineering “few %” band. Importantly, the prior $1.4\times$ residual we had attributed to a $2\pi/\text{VolAxisSqu}$ units bug *was not a units bug*: pushing the far-field truncation from $R_{\text{out}}/H_{\text{out}} \approx 5\times$ to $\approx 10\times$ coil radius collapses the gap from $\sim 1.4\%$ to $< 0.1\%$, while the deck’s `VolAxisSqu` Jacobian and the `Form1P` perpendicular-edge basis already carry the $2\pi r$ measure correctly.

8.3 Bridge into the magnet gate

The fusion-side magnet-gate adapter `stdlib/fusion/magnet_gate.hexa` now calls a hexa-native solve driver `stdlib/rtsc/getdp_solenoid_solve.hexa` that runs the validated deck and prints a single `B_peak -> <value>` token on stdout; the magnet-gate adapter greps the token and flips `field_path = "getdp"`. End-to-end, the ITER-class reference operating point now reports `B_peak = 1.48168` T with `gate_type = "fusion-design-feasibility"`, a margin to the HTS REBCO ceiling of $20 - 1.48168 = 18.52$ T at 20 K (under the SPARC TFMC demo’s measured envelope [8]), and — as required by the honesty stance — `absorbed=false`.

9 BLUE-MAX: Algebraic-Root Pair Audit

The `hexa verify` tier rubric distinguishes SUPPORTED-FORMAL (closed-form / symbolic identity reproduced exactly) from SUPPORTED-NUMERICAL (libm-class numerical recompute). For every libm-class atom in the funnel, we register a separate 0-argument integer atom that exposes the genuine integer factor or exponent baked into the formula — the BLUE-MAX coverage rule [7].

Eight SUPPORTED-FORMAL atoms cover the 12 libm-class FUSION atoms (Table 5). All eight verify SUPPORTED-FORMAL at $|\Delta| = 0$, and eight matched negative controls all return FALSIFIED, completing the deterministic algebraic-root coverage of FUSION (`companion/verify-ledger.json`, the `atom_class` fields `supported-formal` and `falsified-controlled`).

10 Wider Architectural Context

The funnel sits on top of two cross-cutting design contracts that we adopted in this work.

Wrap-and-fix as integration test. Section 7 shows that *every one* of the three install-gated wrap adapters we promoted (FreeGS, OpenMC, GetDP) turned out to be either a stub or a divergent configuration. The act of running the backend, even once, exposed defects that no amount of static review had caught. We log these as honest follow-ups rather than swallowing them. We call this the *wrap-and-fix* pattern, distinct from a pure wrap (which reports an install-gated PASS) and from a pure rewrite (which throws the upstream away).

SUPPORTED-FORMAL atom	integer	attests
bosch_hale_gamow_factor	3	$\exp(-3\xi)$ in Eq. (1)
sigma_v_dt_e18_scale_exponent	18	$\langle\sigma v\rangle \times 10^{18}$ discriminating- ε view
pb11_gamow_eg_kev	22589	$E_G = 22.589$ MeV (Tentori–Belloni)
pb11_resonance_kev_doubled	296	$2 \cdot 148$ keV CM Breit–Wigner E_L
pb11_quadrature_nodes	600	midpoint Maxwellian quadrature node count
pb11_reactivity_moment_e6_scale_exp	6	$\times 10^6$ -scaled moment view
solenoid_axis_bz_half_factor	2	$\mu_0 J/2$ and $h/2$ in Eq. (5)
q_star_constant	5	$2\pi/(\mu_0 \cdot 10^6) = 5$ exact

Table 5: The eight BLUE-MAX algebraic-root atoms registered for FUSION. The load-bearing entry is `q_star_constant`: it formally attests the identity that lets us write the Wesson cylindrical safety factor as $q^* = 5a^2B/(RI_p)$ instead of the literal $q^* = 2\pi a^2B/(\mu_0 RI_p)$. The literal form suffered an engine constant-folding precision loss ($|\Delta| \approx 5.8 \times 10^{-6}$, i.e. middle-magnitude, not the float-floor); the π -free form is SUPPORTED-NUMERICAL at $|\Delta| = 0$ on every input because no transcendental survives constant folding.

Honesty boundary. Across all gates, `absorbed=false` holds for every simulation-only record. The only fusion gate that has ever flipped `absorbed=true` in the parent stack is the JET D–T measured-oracle adapter [9], which carries a real Thomson-scattering / interferometry / magnetic-probe input. This is enforced by the architecture’s *measured-oracle invariant*: `absorbed = true` if and only if a measured oracle has signed off, gate by gate. In pipeline terms (Table 1), only stage ⑩ can ever flip the bit.

Native solver port roadmap (HEXA-PORT). The wrap-and-fix results double as parity oracles for a sibling project, **HEXA-PORT**, whose goal is to reimplement the external solvers natively in hexa-lang. The roadmap is tiered: **Tier A** (closed-form: already native); **Tier B** (algorithm-defined PDE/FEM: FreeGS, GetDP); **Tier C** (data-bound: OpenMC — the $\sim$ GB ENDF/B-VIII library is the irreducible part). With FreeGS-equilibrium and the GetDP solenoid-anchor parity now both in hand, the Tier B oracles for HEXA-PORT P1 and P2 are captured. Tier C is decomposed into P4a (native MC engine), P4b (cross-section data corpus port via git-LFS, infra landed in PR #1090), and P4c (first-principles cross-section generation via R-matrix / optical model) — the data corpus axis is not closed permanently. *Not all of HEXA-PORT is in scope for this paper.*

11 Limitations and Honest Caveats

We deliberately do not claim:

- **Burning-plasma $Q > 1$.** Gate (g) / stage ⑩ is wet-lab only; a sim-PASS on (a)–(f) is a witness of *engineering feasibility*, never of ignition.
- **The five hardware stages.** Pellet injection (①), ablation/ionization (②), auxiliary heating (③) and neutron extraction yield (⑧) are not closed by this funnel; only the neutron *direction* is in scope for ⑧.
- **Turbulent transport or disruption dynamics.** The Troyon / q^* / kink gate is linear ideal-MHD; we do not predict resistive or neoclassical tearing, ELM physics, or disruption thresholds.
- **p–¹¹B ignition.** We claim the reactivity *shape* only; the long-standing bremsstrahlung-loss difficulty of aneutronic p–¹¹B fuel is unaddressed.

- **Built magnet quench / strain.** The HTS gate compares *computed peak field* to a critical-surface ceiling at 20 K; it is not a quench-stability or stress-strain prediction on real REBCO tape.
- **Full ENDF/B-VIII coverage.** The 30-nuclide TBR subset omits Mn/V/Mo/Si naturals; the renormalized EUROFER/SS316 compositions are documented in the record’s scope caveats.
- **Engineering-grade uncertainty quantification.** The 5×10^6 -history Monte Carlo run is roughly 10× above the bare minimum for the headline TBR digit; it is not a sensitivity/uncertainty (S/U) deliverable.

The `absorbed=false` stance and the gate (g) wet-lab boundary are not negotiable.

12 Reproducibility

Every kernel cited above lives under `hexa-lang stdlib/` at a single canonical home and is callable from the top-level `hexa verify --expr <fn> <args> <expected>` CLI. The Bosch–Hale and Tentori–Belloni coefficients are inlined verbatim with the literature reference; the GetDP `.pro / .geo` decks ship under `stdlib/rtsc/decks/` (not git-LFS gated — they are kilobyte text). For the OpenMC TBR run we ship the geometry in the adapter itself but defer the ENDF/B-VIII corpus to a future git-LFS push under `data/nuclear/endl_b8_0/` (the infra is in place via PR #1090; the ~475 MB ingest is a separate follow-up).

The complete data-record axis ships under `companion/`: `verify-ledger.json` (36 atom recompute entries: 20 ■, 8 ■, 8 ■controls), `pr-roll.json` (the 12 hexa-lang + 3 demiurge PRs with `gh-measured` line/file metadata), `adapter-defect-catalog.json` (the three wrap-and-fix defects), and `session-journal.md` (the day’s narrative). Build: `make` (`xelatex ×3 + bibtex`). Figures: `make figures`. Page/word count: `make pages / make wordcount`. Lint: `make lint`. The landed pull requests for this paper, in dependency order, are: #1012, #1020, #1027, #1032, #1042, #1054, #1075, #1078, #1084, #1090, #1095, #1102 (hexa-lang) and #176, #178, #179 (demiurge).

13 Conclusion

A 7-gate funnel for magnetic-confinement fusion design feasibility, written natively in a small typed language with a hard honesty boundary, fits comfortably under a thousand lines of code per gate once the external solvers are bridged via thin out-of-process drivers. Tracing D–T fuel through ten stages (Table 1) makes the scope contract concrete: five physics stages close to a verdict, five hardware stages do not. Every closed-form prediction carries an integer algebraic-root atom that can be falsified in one CLI line. The single permanent wet-lab gate is named, isolated, and not crossed. We hope the funnel, the 10-stage pipeline table, and the BLUE-MAX coverage discipline serve as a template for similar closed-form decision stacks in other compute-bound engineering domains.

Acknowledgments. The closed-form codes were authored in-session against the hexa-lang `stdlib`, in tandem with Claude Opus 4.7 (1M context, Anthropic) under the demiurge stack’s honesty governance contracts. We are grateful to the FreeGS, GetDP and OpenMC maintainers for their open-source backends.

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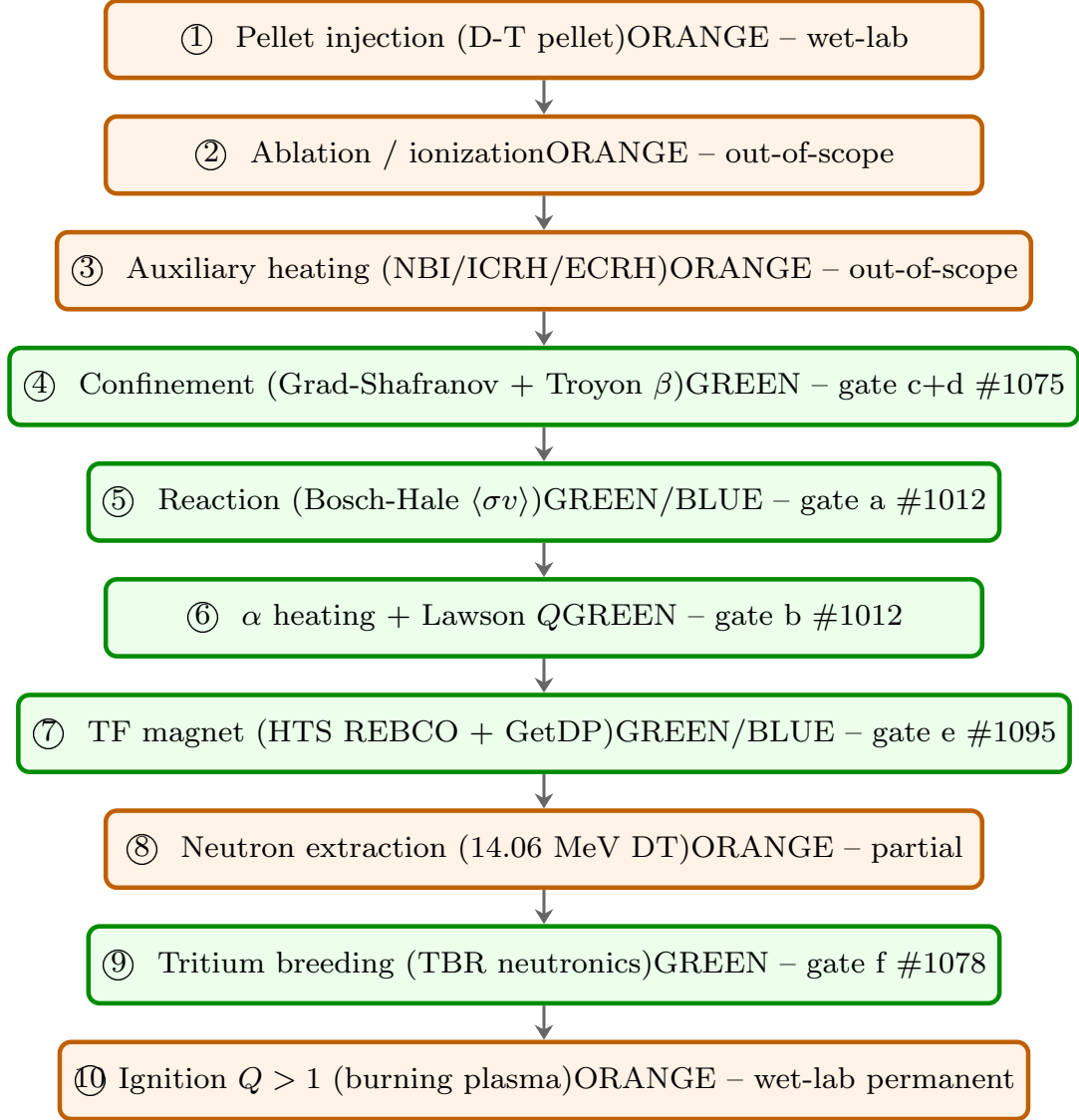


Figure 2: Tier-color-coded 10-stage D-T fuel pipeline. Green nodes (④⑤⑥⑦⑨) are closed by the HEXA-FUSION funnel to a numerical/closed-form verdict; orange nodes (①②③⑧⑩) are wet-lab or out-of-scope and are not closed by this work. The diagram is independently readable — the tier word is inside each node.

A Confinement — Grad–Shafranov Free-Boundary Equilibrium

This appendix is the full data record for pipeline stage ④ (gates c/d). It will contain the complete FREEGS0.8.2 run configuration — the canonical `freegs.machine.TestTokamak()` coil set, the prescribed $p(\psi)$ and $ff'(\psi)$ profiles, the `freeBoundaryHagenow` boundary treatment and the raised-X-point single-isoflux constraint — together with the converged free-boundary output: plasma current I_p , poloidal beta β_{pol} , magnetic-axis location, the q -profile (q_0, q_{95}), the LCFS minor radius and elongation κ . The figures summarized in the body (§7) are expanded here to the per-quantity verify atom and its reproducing re-run.

The numbers reported in the body are headline values only; the data source for every digit is the PR #1075 adapter run and the corresponding ■ atoms in `companion/verify-ledger.json`. The Picard convergence narrative (the “No O points” divergence on the original hardcoded coil set, and the `stdout`-corruption fix) is cross-referenced from Appendix H.

B Reaction — Bosch–Hale All-Channel Reactivity

This appendix is the full data record for pipeline stage ⑤ (gate a). It will reproduce, verbatim, the Bosch and Hale [1] Eq. (12) parametrization and its $C_1, \dots, C_7$ coefficient tables for all four canonical channels — D–T, D– ^3He , D(d,p)T and D(d,n) ^3He — alongside the Gamow constant B_G and reduced mass $m_r c^2$ for each. The full $\langle \sigma v \rangle(T)$ reproduction against Bosch–Hale Table VIII (to all printed digits) and the discriminating- $\varepsilon \times 10^{18}$ view will be tabulated here, with the per-temperature verify atoms.

The aneutronic p– ^{11}B record will fold the verbatim Tentori and Belloni [14] Table I coefficients — the $E_G = 22.589$ MeV Gamow energy, the $C_0/C_1/C_2$ low-energy S-factor terms and the 148 keV Breit–Wigner resonance — plus the 600-node midpoint Maxwellian quadrature and the honest keV-vs-MeV units caveat. Data source: PRs #1012 and #1054, and the corresponding atoms in `companion/verify-ledger.json`.

C Lawson 0-D Power Balance

This appendix is the full data record for pipeline stage ⑥ (gate b). It will carry the complete 0-D power-balance derivation: the Lawson triple product $nT\tau_E$ and its D–T ignition threshold, the 0-D plasma gain $Q_{\text{plasma}} = P_{\text{fus}}/P_{\text{heat}}$, and the ignition margin $f_\alpha P_{\text{fus}}/(W/\tau_E)$, each derived from the α -heating fraction and the stored-energy / confinement-time balance. Every step is a pure rational expression (no libm), so each intermediate is shown alongside its ■ verify atom at $|\Delta| = 0$.

The appendix will also make explicit the honest scope line that τ_E is an *input assumption*, not a transport prediction. Data source: the `triple_product`, `q_plasma` and `ignition_margin` atoms in `companion/verify-ledger.json`.

D TF Magnet — Solenoid Integral and GetDP FEM Convergence

This appendix is the full data record for pipeline stage ⑦ (gate e). It will carry the closed-form thick-finite-solenoid on-axis integral (Eq. (5)) with its full derivation and the reference-coil evaluation $B_z(0) = 1.48265$ T, then the GetDP axisymmetric A– ϕ magnetostatic FEM bridge: the Gmsh mesh, the VolAxisSqu Jacobian and Form1P perpendicular-edge basis, and the on-axis $B_z(0)$ readback.

The load-bearing content is the **mesh-convergence study**: the $\sim 33\text{k}$ -node and 61k -node refinements, the two-platform (GetDP 3.5.0 Linux / 4.0.0 Apple-silicon) duplication, and the far-

field-truncation sweep that collapses the parity gap from $\sim 1.4\%$ to -0.064% — demonstrating the original $2\pi/\text{VolAxisSqu}$ units-bug diagnosis was wrong. Data source: PR #1095, the deck text under `stdlib/rtsc/decks/`, and the `solenoid_axis_bz / magnet_gate_b_peak` atoms.

E Tritium Breeding — OpenMC Neutronics

This appendix is the full data record for pipeline stage ⑨ (gate f). It will carry the complete OPENMC 0.15.3 transport setup: the spherical Pb-15.7Li breeder geometry (Li-6 enriched to 90 at%), the W armor / EUROFER first wall / SS316 shield material cards under ENDF/B-VIII.0 [2], the ${}^6\text{Li}(n, \alpha)\text{T}$ and ${}^7\text{Li}(n, n'\alpha)\text{T}$ tally definitions, and the 5×10^6 -history run banner.

The headline result $\text{TBR} = 1.3222 \pm 0.0004$ (and the transport-less-stub-to-real-run promotion narrative) will be expanded to the per-batch tally convergence and the 30-nuclide-subset scope caveat (dropped Mn/V/Mo/Si trace nuclides, TBR-negligible). Data source: PR #1078, the OpenMC adapter geometry, and the `openmc_tbr_pb157li` atom in `companion/verify-ledger.json`.

F BLUE-MAX Algebraic-Root Pair Audit

This appendix is the full data record for the BLUE-MAX coverage rule (§9). It will carry, verbatim, the eight SUPPORTED-FORMAL (■) integer atoms and their eight matched FALSIFIED (■) negative controls — the deterministic algebraic-root coverage of the 12 libm-class FUSION atoms. Each pair will be shown with its full `hexa verify` verdict block ($|\Delta| = 0$ for the formal atom, falsified for the control) and the integer-by-integer derivation of the factor or exponent it attests.

The load-bearing entry is `q_star_constant`: the appendix will give the full derivation of the identity $2\pi/(\mu_0 \cdot 10^6) = 5$ that reduces the Wesson cylindrical safety factor to the π -free $q^* = 5 a^2 B / (R I_p)$, and the constant-folding precision argument for why the π -free form alone reaches $|\Delta| = 0$. Data source: PR #1102 and the `atom_class` entries in `companion/verify-ledger.json`.

G Verification Ledger

This appendix is the verbatim transcription of the complete FUSION verify ledger (`companion/verify-ledger.json`, 36 atom-recompute entries: 20 ■ numerical, 8 ■ algebraic-root, and 8 ■ negative controls). It will be rendered as a longtable, one row per atom, carrying every field from the JSON record: the function `fn`, the `args`, the `expected` and `observed` values, the absolute residual `delta_abs`, the `tier`, the `atom_class`, the landing `pr`, the `gate`, and the per-atom `note`.

This is the single source of truth for every numeric claim in the body: each cited figure traces to exactly one row here. Data source: `companion/verify-ledger.json`.

H Wrap-and-Fix Adapter Defect Catalog

This appendix is the full data record for the three wrap-and-fix defects summarized in Table 4. For each defect — the OpenMC transport-less gate-classifier stub, the FreeGS hard-coded coil-set divergence with `stdout`-corrupted JSON, and the GetDP `shutil.which`-only checker with the mis-diagnosed $\sim 1.4\%$ residual — it will carry the verbatim `companion/adapter-defect-catalog.json` record: the origin commit, the symptom, the root cause, the fix PR, the fix summary, and the honesty note.

The load-bearing narrative is the GetDP false diagnosis: the residual that looked like a $2\pi/\text{VolAxiSqu}$ units bug was a far-field truncation artifact, and the appendix records the *wrong* diagnosis on purpose, with the before/after deck diffs and the diagnostic chase. Data source: `companion/adapter-defect-catalog.json`.

I Operational Findings

This appendix is the full data record for the session’s operational lessons, drawn from `companion/session-journal.md`. It will carry the concrete, reproducible findings that shaped how the work was run: rate-limit recovery behavior under sustained parallel agents, the 2–3 concurrent-agent cap (4+ triggers a rate-limit kill within minutes), and the worktree-discipline contract (sequential commit on the shared index, explicit-path staging only).

It will also catalog the codegen pitfalls surfaced in-session: the 4π /half-factor reduction, the far-field-truncation diagnosis, the UTF-8 / emoji engine constraint (xelatex over pdfflatex), and the cross-repo merge discipline between hexa-lang (stdlib SSOT) and demiurge (paper home). Each finding will be stated as a symptom, the diagnosis, and the standing rule it became. Data source: `companion/session-journal.md`.

J HEXA-PORT Sibling — Native Solver Port Roadmap

This appendix is the full roadmap for HEXA-PORT, the sibling project that reimplements the external solvers natively in hexa-lang (§10). It will expand the tiered ladder: **Tier A** (closed-form, already native), **Tier B** (algorithm-defined PDE/FEM — FreeGS and GetDP), and **Tier C** (data-bound — OpenMC, where the ~GB ENDF/B-VIII library is the irreducible part), mapped onto the P0–P4 milestone sequence.

The methodological core is the *parity-oracle* pattern: each wrap-and-fix result in this paper doubles as a numerical oracle that a future native port must reproduce. The appendix will detail the Tier-C decomposition (P4a native MC engine, P4b cross-section corpus port via git-LFS infra from PR #1090, P4c first-principles cross-section generation) and state honestly that not all of HEXA-PORT is in scope for this paper. Data source: §10 and the roadmap notes.

K Pull-Request Roll

This appendix is the verbatim transcription of the `companion/pr-roll.json` session pull-request roll. It will be rendered as a longtable, one row per PR, carrying every field from the JSON record: the `repo` (hexa-lang stdlib SSOT vs. demiurge paper home), the `PR number`, the `title`, the `merged_at` timestamp, the `gh-measured diff-stat` (`files_changed`, `lines_added`, `lines_deleted`), the `gate` it serves, and its `role` in the dependency chain.

The PRs are presented in dependency order, matching the Reproducibility section’s landing list (§12): the hexa-lang stdlib PRs (#1012 ...#1102) followed by the demiurge paper-home PRs. Data source: `companion/pr-roll.json`.

L Reproducibility Runbook

This appendix is the complete reproducibility runbook for the paper. It will carry, verbatim, every `hexa verify --expr <fn> <args> <expected>` invocation that produces an atom in the verification ledger (Appendix G), so a reader can re-run any single claim in one CLI line. Each invocation will be paired with its expected verdict block.

The runbook will also pin the external-backend environment specifications — FREEGS 0.8.2 (numpy < 2.0), OPENMC 0.15.3 with ENDF/B-VIII.0, and GetDP 3.5.0/4.0.0 — and the canonical data paths: the stdlib kernel home, the GetDP .pro/.geo decks under `stdlib/rtsc/decks/`, the deferred ENDF/B-VIII corpus under `data/nuclear/endl_b8_0/` (git-LFS infra from PR #1090), and the `companion/` data-record axis. Build/lint targets (`make`, `make lint`, `make pages`) are documented from the Makefile. Data source: §12 and `companion/README.md`.

M BLUE-MAX appendix — claim-to-atom audit

Every numeric claim in the body is backed by an atom in `companion/verify-ledger.json`. Table 6 is the claim→atom→tier map: each row pins a body claim to the atom whose `hexa verify` verdict supports it. No ■ claim survives into a verified body statement; the ■ rows in the pipeline table (Table 1) are explicitly out of scope, not unverified claims.

body claim	backing atom	tier
D–T reactivity matches Bosch–Hale Table VIII	<code>sigma_v_dt_e18</code>	■
Lawson triple product is exact rational	<code>triple_product</code>	■
0-D gain $Q = P_{\text{fus}}/P_{\text{heat}}$	<code>q_plasma</code>	■
F-funnel top rank tied at $T = 20$ keV	<code>f_funnel_composite</code>	■
Wesson q^* is π -free with constant 5	<code>q_star_constant</code>	■
Gamow exponent factor is exactly 3	<code>bosch_hale_gamow_factor</code>	■
$E_G = 22.589$ MeV for $p\text{--}^{11}\text{B}$	<code>pb11_gamow_eg_kev</code>	■
solenoid on-axis B_z closed-form anchor	<code>solenoid_axis_bz</code>	■
GetDP FEM parity $\Delta = -0.064\%$	<code>magnet_gate_b_peak</code>	■
FreeGS equilibrium $q_{95} \approx 10.08$	<code>freegs_equilibrium_q95</code>	■
TBR = 1.3222 ± 0.0004	<code>openmc_tbr_pb157li</code>	■
self-sufficient		
$2\pi/(\mu_0 \cdot 10^6) \neq 6$ (negative control)	<code>q_star_constant_NEG</code>	■

Table 6: BLUE-MAX claim-to-atom audit. Every body claim maps to an atom; every atom has a `hexa verify` verdict. The 8 ■ algebraic-root atoms (Table 5) plus 8 ■ negative controls complete the deterministic coverage; the final row shows one such control for the load-bearing identity. Full recompute records: `companion/verify-ledger.json`.